

GeoROS++: A Georeferenced Real-Time Stitching Method for Sparse Aerial Orthophotos

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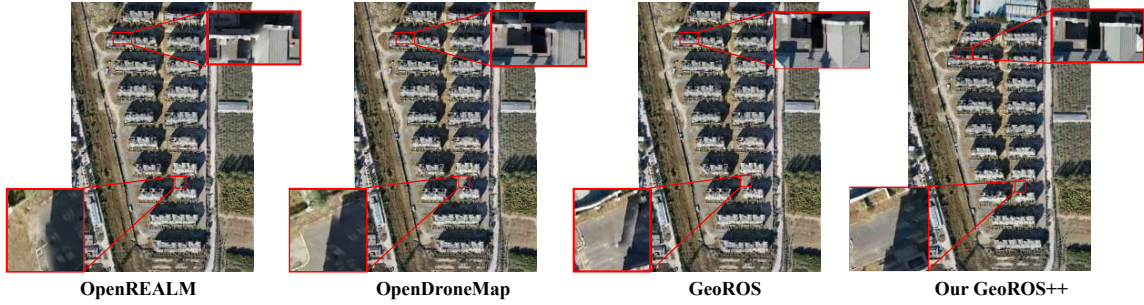


Figure 1: Orthomap comparison on the City dataset. Our GeoROS++ achieves the most accurate geometry and visual coherence, especially around buildings, compare to existing method such as OpenREALM [Kern and Bobbe 2020], OpenDroneMap[OpenDroneMap 2025], and GeoROS [Gao et al. 2022].

Abstract

Real-time alignment of large-scale, high-resolution aerial imagery is a core challenge in sequential estimation for dynamic environments. Existing methods often fail or lose accuracy when faced with large viewpoint changes (i.e., wide baselines) or strong illumination variations. We propose **GeoROS++**, an adaptive framework for real-time geometric estimation. Its core is an *Adaptive Multi-motion model*

Selection (AdMS) strategy that selects the best geometric model online using real-time data statistics. To improve robustness, we incorporate an online *Illumination Compensation Module (ICM)* that enhances feature invariance under lighting changes. We also adopt GPU-accelerated RootSIFT for fast and reliable low-level feature extraction. Experiments on both public and self-collected challenging datasets validate that our GeoROS++ outperforms existing methods in geometric accuracy and real-time performance.

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CCS Concepts

• Computing methodologies → Image processing.

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1 INTRODUCTION

Orthophoto stitching from UAV imagery has been rapidly adopted across various industries due to its low cost and high efficiency. However, achieving consistent stitching quality while maintaining low computational complexity and high accuracy remains challenging. Structure-from-Motion (SfM) methods can deliver high accuracy and stability by reconstructing the 3D structure of scenes, but are computationally expensive. Recent learning-based methods have further advanced high-fidelity offline reconstruction by generating detailed 3D terrain models from aerial imagery [Feng and Atanasov 2024]. In contrast, visual SLAM approaches focus on estimating camera poses and generating sparse 3D point clouds in real time, trading some accuracy for higher efficiency [Dwimantara et al. 2025].

Incorporating geographic information into orthophoto stitching has shown promise, as GPS-based motion models can accelerate optimization convergence and reduce the likelihood of relocation, while robust GPS constraints during mapping increase tolerance to positioning errors. Such designs have demonstrated strong accuracy and efficiency in both public and self-collected datasets. As illustrated in Figure 1, incorporating geographic information significantly improves orthomap consistency and visual fidelity.

Nevertheless, as orthophoto stitching is deployed in broader applications, the need for both robustness and efficiency becomes more critical. With the availability of high-performance hardware such as GPUs, descriptors like RootSIFT [Arandjelović and Zisserman 2012] should be prioritized. At the same time, data complexity and robustness requirements in tracking are increasing. For example, *large-interval data* makes feature matching significantly harder: local search may fail to find correspondences, while global search may yield ambiguous matches. In addition, insufficient or uneven lighting in collected images poses another challenge. Existing approaches often fail under these conditions, leading to stitching errors or incomplete results.

Our *GeoROS++* is based on GeoROS [Gao et al. 2022], which is an enhanced UAV orthophoto stitching framework with improved robustness, efficiency, and adaptability. The main contributions are:

- **GPU-accelerated RootSIFT feature extraction:** Leveraging GPU computing power, RootSIFT offers greater robustness to photometric variations.
- **Illumination Compensation Module (ICM):** A novel photometric normalization method that adaptively corrects brightness and contrast.
- **Adaptive Multi-motion model Selection (AdMS):** An effective GPS-based small-rotation motion model, combined with a new adaptive strategy.

2 METHODOLOGY

In GeoROS [Gao et al. 2022], the georeferenced tracking module is the foundational stage of the real-time orthophoto stitching pipeline, directly influencing the quality of subsequent mapping

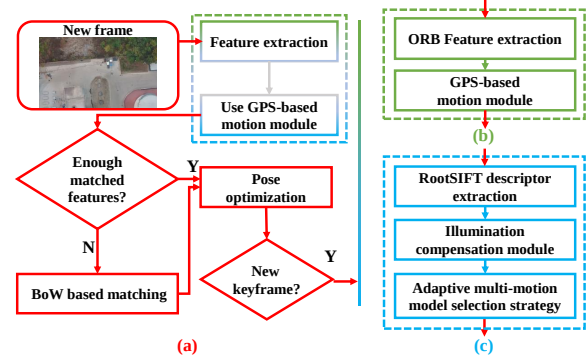


Figure 2: Comparison of GeoROS vs. our GeoROS++: (a) Tracking overview; (b) GeoROS modules; (c) Enhanced modules and ICM in our GeoROS++.

and stitching. Our GeoROS++ framework is designed to enhance the performance of this module in terms of both speed and robustness.

Figure 2 illustrates the differences in the georeferenced tracking pipeline between GeoROS and our proposed GeoROS++, with our main contributions highlighted in dotted boxes. We replace the ORB descriptor with RootSIFT, which can be more efficiently accelerated on the GPU. Additionally, we employ a more adaptable motion model selection strategy to handle aerial imagery from complex scenarios.

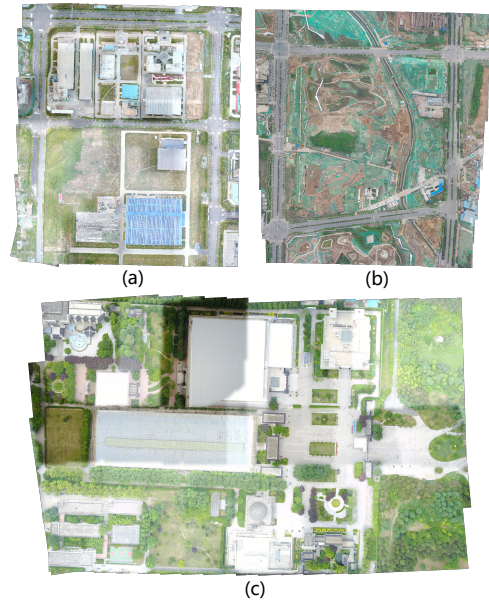


Figure 3: Orthophoto mosaics by our GeoROS++ on *mavic-factory* (a), *mavic2-road* (b), and *mavic-warriors* (c). These datasets have large inter-frame baselines (~500 px vs. ~60 px normally), where GeoROS fails to stitch, but GeoROS++ produces high-quality, geometrically consistent mosaics.

2.1 RootSIFT Descriptor Extraction

GeoROS++ replaces ORB [Rubblee et al. 2011] with RootSIFT [Arandjelović and Zisserman 2012], which is more robust to photometric

and geometric variations while remaining GPU-efficient. Root-SIFT L^1 -normalizes each SIFT [Lowe 1999] descriptor and applies an element-wise square root, enabling the Hellinger kernel: $H(x, y) = \sum_{i=1}^n \sqrt{x_i y_i} = 1 - \frac{1}{2} \|\sqrt{x} - \sqrt{y}\|_2^2$, where x, y are normalized histograms.

2.2 Illumination Compensation Module

After RootSIFT extraction, we apply an *Illumination Compensation Module* (ICM) to normalize photometric variations between consecutive frames, mitigating feature matching degradation under lighting changes. Let $\mathcal{H}_t = \{I_{t-k}\}_{k=1}^{K_t}$ be the set of matched historical frames before time t , with K_t determined by available matches. For each $I_k \in \mathcal{H}_t$, the mean grayscale intensity is $\mu_k = \frac{1}{|\Omega|} \sum_{(u,v) \in \Omega} I_k(u, v)$, and the weighted historical reference intensity is $\bar{\mu}_t = \frac{\sum_k w_k \mu_k}{\sum_k w_k}$, where w_k is proportional to the inlier matches between I_k and I_t . For a new frame I_{t+1} with mean μ_{t+1} , we compute $I'_{t+1}(u, v) = \alpha_{t+1} I_{t+1}(u, v) + \beta_{t+1}$, $\alpha_{t+1} = \frac{\sigma_{\bar{\mu}_t}}{\sigma_{\mu_{t+1}}}$, $\beta_{t+1} = \bar{\mu}_t - \alpha_{t+1} \mu_{t+1}$, with $\sigma_{\bar{\mu}_t}$ and $\sigma_{\mu_{t+1}}$ the intensity standard deviations of the reference and current frames. This on-the-fly normalization aligns brightness, yielding more uniform mosaics under heterogeneous lighting (Figure 3).

2.3 Adaptive Multi-motion model Selection

Adaptive Multi-motion model Selection (AdMS) dynamically selects the most suitable motion model using scene geometry and GPS priors. It is particularly effective for *small-rotation* cases, reducible from full 3D to efficient 2D estimation.

Small-rotation GPS model. For consecutive frames F_{t-1} and F_t at altitude h with small yaw θ ($\|\theta\| \ll 1$), let $\mathbf{g}_{t-1}, \mathbf{g}_t \in \mathbb{R}^2$ be planar GPS coordinates. The displacement is: $\Delta \mathbf{g}_t = \mathbf{g}_t - \mathbf{g}_{t-1}$. Given inlier matches $\mathcal{M}_{t-1,t} = \{(\mathbf{p}_i^{t-1}, \mathbf{p}_i^t)\}_{i=1}^{N_m}$, the image-plane displacements are: $\mathbf{d}_i = \mathbf{p}_i^t - \mathbf{p}_i^{t-1}$. We fit a similarity transform: $\mathbf{p}_i^t \approx \hat{\mathbf{s}} \mathbf{R}(\hat{\theta}) \mathbf{p}_i^{t-1} + \hat{\mathbf{t}}$, via least squares: $\min_{\hat{\mathbf{s}}, \hat{\theta}, \hat{\mathbf{t}}} \sum_{i=1}^{N_m} \|\mathbf{p}_i^t - \hat{\mathbf{s}} \mathbf{R}(\hat{\theta}) \mathbf{p}_i^{t-1} - \hat{\mathbf{t}}\|_2^2$.

Keypoints $\mathcal{K}_{t-1}$ are predicted in F_t as: $\hat{\mathbf{p}}_j^t = \hat{\mathbf{s}} \mathbf{R}(\hat{\theta}) \mathbf{p}_j^{t-1} + \hat{\mathbf{t}}$, and matched within local windows to reduce search cost and mismatches. AdMS applies: (1) GPS-based tracking (full 3D if available, else 2D) + standard tracking; (2) if failed and $\|\hat{\theta}\|$ is below an MAD-based threshold, invoke the small-rotation GPS model; (3) otherwise, fallback to image-only bag-of-words, then brute-force matching. All thresholds are computed from historical match statistics, avoiding manual tuning.

3 EXPERIMENTS AND RESULTS

We conducted comprehensive experiments to evaluate the effectiveness of GeoROS++. All evaluations were performed on a workstation equipped with an Intel i7-9700 CPU, 32 GB RAM, and an NVIDIA GeForce RTX 2060 GPU with 12 GB VRAM.

3.1 Dataset

Table 1 summarizes all datasets. We collected three of them (lawn, urban, construction) using a DJI Mavic2 with GPS. The lawn dataset includes four chessboard GCPs (2×2 and 4×4 grids, each 0.5m or

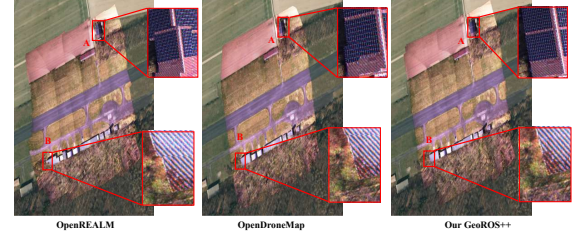


Figure 4: Orthomap comparison on the edm dataset. Our GeoROS++ preserves better structural details and avoids grid distortions seen in other methods.

0.25m wide). All datasets use standard GPS with decimeter-level horizontal accuracy and up to meter-level max error.

The first three datasets have normal image spacing, highlighting GeoROS++ precision. Later datasets have wider spacing and challenging conditions (e.g., uneven lighting), where original GeoROS fails (Figure 3), showing GeoROS++ robustness.

3.2 Visual Comparison

We compare OpenREALM [Kern and Bobbe 2020], OpenDroneMap [OpenDroneMap 2025], GeoROS [Gao et al. 2022], and our GeoROS++ on two representative datasets. In the urban scene (Figure 1), OpenREALM and OpenDroneMap exhibit noticeable geometric distortions around buildings and road boundaries, as highlighted in the zoomed regions. GeoROS achieves better alignment but still introduces slight misalignments. Our GeoROS++ delivers the most geometrically accurate and visually coherent orthomap. In the solar panel scene (Figure 4), OpenREALM and OpenDroneMap produce visible artifacts and grid distortions. In contrast, GeoROS++ preserves structural consistency and fine details, yielding the most reliable mapping quality.

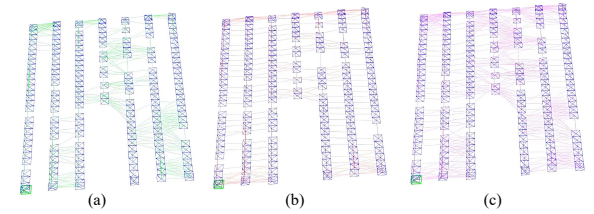


Figure 5: Different forms of local maps: (a) based on common view relationships, (b) based on GPS information, and (c) based on both visual and geographic information, which has more reliable pose estimation and orthomosaic alignment.

3.3 Local Mapping

Integrating visual and GPS information enhances local map efficiency for exploring nearby areas. As shown in Figure 5, all maps use 9 neighboring keyframes. The covisibility-only map (a) fails to capture inter-route relationships. The GPS-only map (b) captures these better but overlooks keyframes rich in visual landmarks. The fused map (c) combines GPS efficiency with visual accuracy. Thus, our method adopts this joint formulation for all local maps.

Table 1: Parameters of datasets used in the experiments.

Dataset	Altitude (m)	Area (km ²)	Resolution	Images
lawn	80.1	0.036	1920×1080	817
urban	99.7	0.166	1920×1080	863
construction	100.1	0.098	1920×1080	830
edm	99.5	0.106	1228×1027	3276
phantom-village	196	0.932	1920×1080	761
phantom-olathe	122	0.257	4000×3000	160
mavic-road	205	0.568	5472×3078	240
mavic-campus	213	0.772	4000×3000	293
mavic-factory	153	0.459	4000×3000	359
mavic-huangqi	188	0.352	4000×3000	229
mavic-warriors	143	0.157	4000×3000	96

3.4 Effect of GPS on Mapping Accuracy

Figure 6 compares sparse maps with (b) and without (a) GPS. UAV altitude drift is minimal in practice. The GPS-integrated system better aligns with the true trajectory and preserves a consistent ground plane, while the non-GPS version suffers from drift and distortion, reducing mapping accuracy.

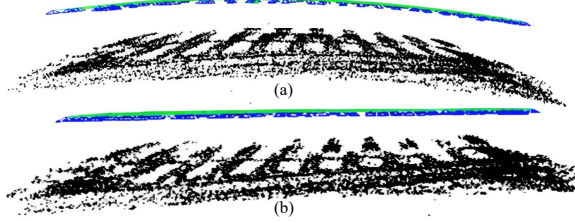


Figure 6: Sparse reconstruction results of the system: (a) Without GPS, the map shows distorted trajectories and warped structures; (b) With GPS, the map becomes smoother and more consistent.

Table 2: Runtimes (in seconds) of different methods on four datasets.

	Lawn	City	Edm	Pv3
openDroneMap	8361	8906	30017	7393
DJI Terra	693.96	608	7357	308
OpenREALM	62	106	493	491
GeoROS	67	116	262	114
Our GeoROS++	61.2	108	257	109

3.5 Comparison on Runtime and GCP Accuracy

All methods were tested on an Intel i7-9700 CPU with 32GB RAM. As shown in Table 2, our GeoROS++ runs fastest. Table 3 shows GCP errors on the *lawn* dataset, where GCPs are checkerboard centers and error is the GPS-to-ground-truth distance. GeoROS++ achieves the lowest error, demonstrating effective GPS utilization.

Table 3: Ground Control Point (GCP) errors (in meters) for different methods.

	OpenREALM	DJI Terra	openDroneMap	GeoROS	Our GeoROS++
GCP1	11.7	36.8	39.3	2.6	2.5
GCP2	9.9	37.2	38.8	2.8	2.6
GCP3	6.0	36.4	34.1	1.4	1.3
GCP4	8.5	35.1	33.3	1.4	1.2
Aver	9.0	36.3	36.4	2.1	1.9

Table 4: Performance comparison of GeoROS++ variants with different components. GE: GCP Error (m); RT: Runtime (s); MS: Match Success (%).

Variant	RootSIFT	ICM	AdMS	GE(m)	RT(s)	MS (%)
Our GeoROS++	✓	✓	✓	1.9	61.2	98.3
w/o RootSIFT	×	✓	✓	3.5	68.4	91.7
w/o ICM	✓	×	✓	2.8	61.0	94.1
w/o AdMS	✓	✓	×	3.1	60.7	89.4
GeoROS	×	×	×	2.1	67.0	92.5

3.6 Ablation Study on Core Modules

We ablate GeoROS++ by removing one core module at a time while keeping other settings fixed (Table 4). Evaluated metrics are: (i) GCP error on *lawn* ($\downarrow$), (ii) runtime ($\downarrow$), and (iii) match success rate ($\uparrow$). Variants include: (1) *w/o RootSIFT* (use ORB), (2) *w/o ICM* (no illumination compensation), (3) *w/o AdMS* (no adaptive motion model), and (4) *Original GeoROS* (no proposed modules).

4 CONCLUSION

We propose GeoROS++, a georeferenced real-time robust orthophoto stitching framework that addresses degradation in sparse aerial imagery. Our GeoROS++ improves both real-time performance and robustness on public and self-collected datasets. GPU-accelerated RootSIFT descriptors enhance feature extraction. The Illumination Compensation Module (ICM) mitigates uneven lighting effects, while the Adaptive Multi-motion model Selection (AdMS) strategy effectively handles large-interval data. Compared to state-of-the-art methods, our GeoROS++ offers superior adaptability and accuracy.

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